

Cross-Currency Dynamics in Cryptocurrency Spot Markets under Stablecoin Stress

Evidence from the March 2023 USDC De-pegging Episode

IAQF 2026 Student Competition Submission

1. Research Questions & Hypotheses

This study examines how Bitcoin’s price, liquidity, and microstructure differ across quote currencies—flat USD versus stablecoins USDT and USDC—and how stablecoin credibility shocks reshape cross-venue dynamics. The analysis centers on the 21-day window of 1–21 March 2023 (UTC), which includes the USDC de-pegging episode triggered by the Silicon Valley Bank failure.

For the cross-currency basis component (Q1), we test the following hypothesis:

- **H1 (Cross-currency basis).** A persistent, economically meaningful basis exists between BTCUSDT and BTCUSD after transaction costs. This basis is directional (USDT discount), widens under stablecoin stress, and reverts slowly, reflecting funding-risk premia and limits to cross-venue arbitrage.

2. Data & Methodology

2.1 Data sources and sample

We collect 1-minute OHLCV candle data for the period 1–21 March 2023 (UTC) from two major spot exchanges:

- **Binance:** BTCUSDT (21 daily CSV files via public kline archive).
- **Coinbase:** BTCUSD (REST API candles endpoint, aggregated into a single file).

Binance trades nearly continuously, producing $\sim 30,240$ candles over 21 days. Coinbase exhibits occasional genuine no-trade minutes reflecting thinner on-shore USD liquidity. After inner-joining on a common 1-minute UTC grid and removing candles where either venue reports no trade, the aligned sample contains $N \approx 29,950$ observations.

2.2 Basis construction

Let P_t^{USDT} and P_t^{USD} denote the 1-minute close prices on Binance and Coinbase, respectively. The log cross-currency basis is

$$b_t = \log P_t^{\text{USDT}} - \log P_t^{\text{USD}}.$$

A negative basis indicates that BTC is cheaper when priced in USDT than in USD.

2.3 Transaction-cost band

We define a no-arbitrage band based on spot taker fees: 0.10% (Binance) + 0.40% (Coinbase), giving a symmetric round-trip cost $\tau = 0.50\%$. Deviations within $\pm\tau$ are treated as economically negligible. Robustness checks vary $\tau \in \{0.30\%, 0.50\%, 0.70\%\}$.

2.4 Regime definition

We define the *stress regime* as March 10–13, 2023 (UTC), corresponding to the SVB failure announcement through the peak of the USDC de-peg. All remaining dates constitute the *normal regime*. A binary indicator D_t^{stress} takes value 1 during the stress window.

2.5 Econometric methods

- **Persistence.** AR(1) on b_t for overall and regime-specific half-lives; z -test for equality of ϕ across regimes.
- **Stationarity.** ADF and KPSS tests on the basis series.
- **Second-moment tests.** Levene test for equal variances across regimes; ARCH-LM test for volatility clustering; interaction regression ($D_t^{\text{stress}} \times b_{t-1}$) to test whether ϕ itself shifts.
- **Driver regression.** OLS with HAC (Newey–West, 60 lags) standard errors: b_t on stress dummy, 60-minute realized volatility, scaled volume, and b_{t-1} .
- **Cointegration.** Johansen trace test and Engle–Granger residual-based test on log prices.
- **Price discovery.** VECM with cointegration rank 1; Hasbrouck (1995) information shares from both Cholesky orderings; Granger causality tests (lags 1–5).

3. Empirical Results

3.1 BTC/USD microstructure analysis

[To be completed by team.]

3.2 BTC/stablecoin microstructure analysis

[To be completed by team.]

3.3 Cross-currency basis analysis

3.3.1 Level dynamics and economic significance

Over the full sample, the mean basis is -24 bps ($\sigma = 29$ bps), indicating a small but systematic USDT discount. During the March 10–13 stress episode, the discount widens to roughly -100 to -150 bps before settling into a more negative post-stress plateau (Figure 1). Approximately 13.3% of minutes exceed the $\pm 0.5\%$ no-arbitrage band, and every such exceedance is negative—there are zero positive breaches. This strict one-sidedness rules out microstructure noise and points to a directional funding premium.

An AR(1) specification yields $\hat{\phi} = 0.9985$ (half-life ≈ 448 minutes ≈ 7.5 hours). Once a cost-exceeding dislocation opens, it decays over hours rather than minutes, consistent with limits to arbitrage across fragmented venues.

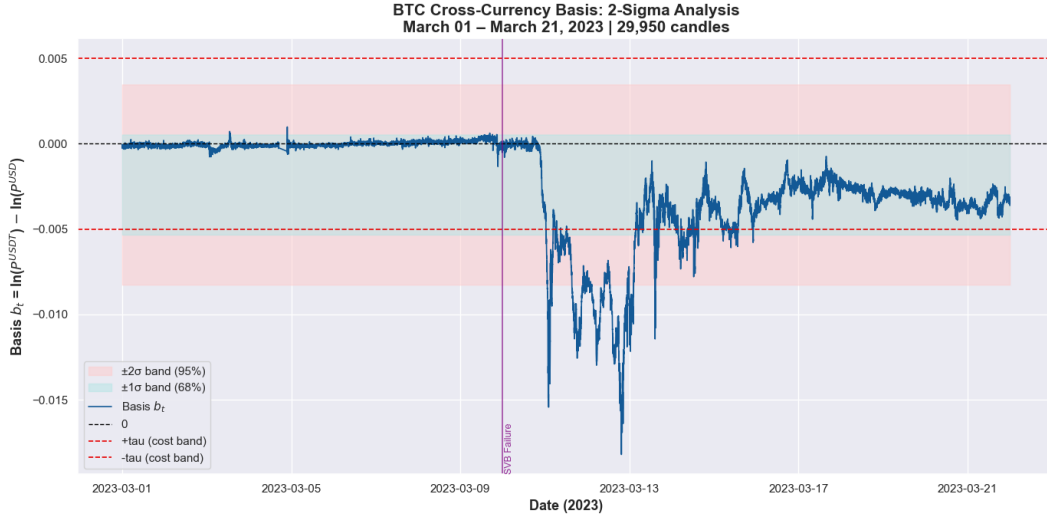


Figure 1: BTCUSDT–BTCUSD log basis (1-minute) with $\pm 0.5\%$ transaction-cost band. Cost-adjusted deviations are concentrated during March 10–13 and are exclusively one-sided (USDT discount).

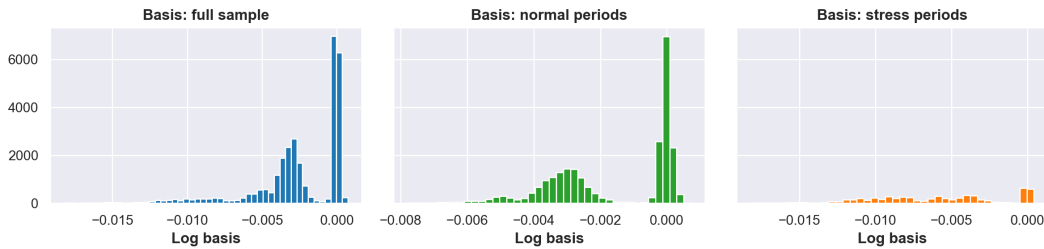


Figure 2: Distribution of the basis in normal vs. stress periods. The stress distribution shifts left with a heavy negative tail, consistent with a persistent USDT discount.

3.3.2 Second-moment analysis: variance regime shift

The stress episode affects the *dispersion* of the basis far more than its *persistence*. The variance ratio of basis innovations (stress/normal) is $5.6\times$, and a Levene test strongly rejects equal variances ($p < 10^{-4}$). An ARCH-LM test on AR(1) residuals confirms pronounced volatility clustering ($p < 10^{-4}$).

Critically, an interaction regression of the form

$$b_t = c + \phi b_{t-1} + \beta_1 D_t^{\text{stress}} + \beta_2 (D_t^{\text{stress}} \times b_{t-1}) + \varepsilon_t$$

shows that the interaction term β_2 is insignificant ($p = 0.38$; HAC standard errors). The AR(1) coefficient

is $\hat{\phi} = 0.9972$ in normal periods and 0.9980 during stress—a difference that is statistically indistinguishable (z -test $p = 0.42$). The arbitrage mechanism persists at the same rate across regimes; only the magnitude of dislocations widens under stress.

3.3.3 Drivers of basis variation

We estimate a reduced-form regression of b_t on a stress indicator, 60-minute realized volatility, scaled USD-venue volume, and the lagged basis, using HAC (Newey–West, 60 lags) standard errors. The stress dummy is the only economically and statistically significant driver beyond the lag ($t = -4.0$, $p < 0.001$), indicating a more negative basis during March 10–13. Realized volatility and volume are not individually significant under HAC inference ($p = 0.15$ and $p = 0.51$), suggesting that the stress effect operates through a broad regime shift rather than through volatility or liquidity channels individually.

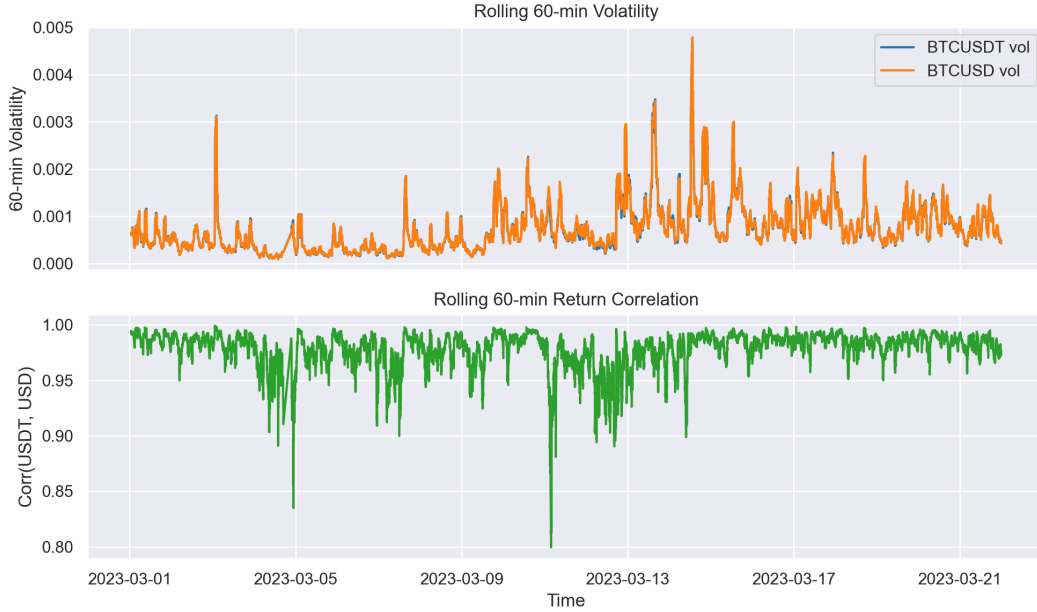


Figure 3: Rolling 60-minute volatility and cross-venue correlation. Volatility spikes sharply on March 10–11 while correlation temporarily drops, consistent with a regime shift in which venues briefly decouple before re-synchronizing.

3.3.4 Cointegration and price discovery

Johansen’s trace test rejects the null of no cointegration at the 1% level (eigenvalue 26.75 vs. 5% critical value 15.49), confirming a long-run equilibrium between the log-price series. Engle–Granger yields $p = 0.18$, which is consistent with its lower power in near-unit-root settings with structural shifts.

Granger causality tests indicate bidirectional information flow, with Binance leading Coinbase more strongly ($p < 10^{-15}$ at lag 2) than the reverse ($p \approx 0.001$).

A Vector Error Correction Model yields error-correction speeds $\hat{\alpha}_{\text{USD}} = -0.0064$ and $\hat{\alpha}_{\text{USD}} = -0.0052$. Coinbase adjusts less rapidly to cointegrating disequilibria, acting as the marginal price leader. Hasbrouck (1995) information shares assign 60.2% (midpoint) of efficient-price variance to Binance and 39.8% to Coinbase, reflecting Binance’s larger volume but Coinbase’s role as the fiat-anchored reference price.

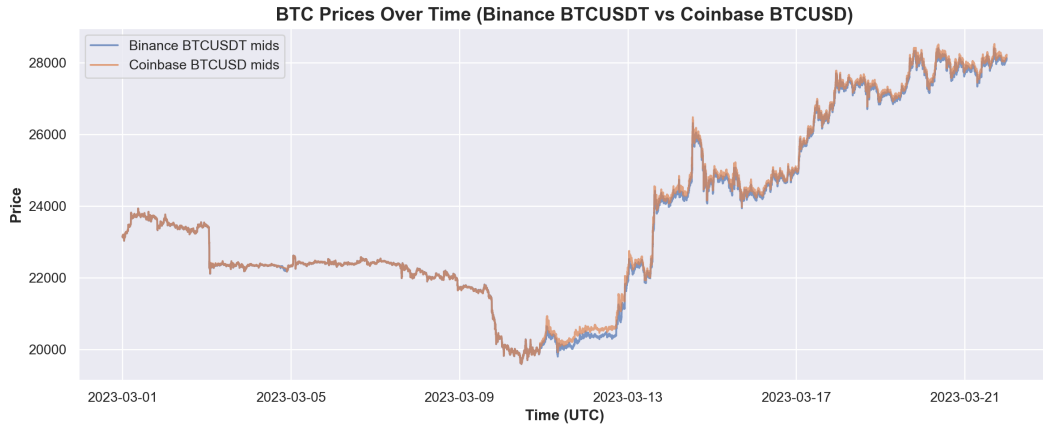


Figure 4: BTCUSDT and BTCUSD mid prices (1-minute). Tight co-movement confirms that basis dynamics reflect funding frictions, not a breakdown of price linkage.

3.3.5 Robustness

Results are robust to two perturbations. First, fee-band sensitivity analysis shows that exceedance shares range from 36.4% ($\tau = 0.30\%$) to 7.9% ($\tau = 0.70\%$), confirming that economically meaningful dislocations are not artefacts of a particular cost assumption. Second, using mid prices instead of close prices yields essentially identical results (mean basis -23.9 bps vs. -24.0 bps; exceedance 13.24% vs. 13.31%).

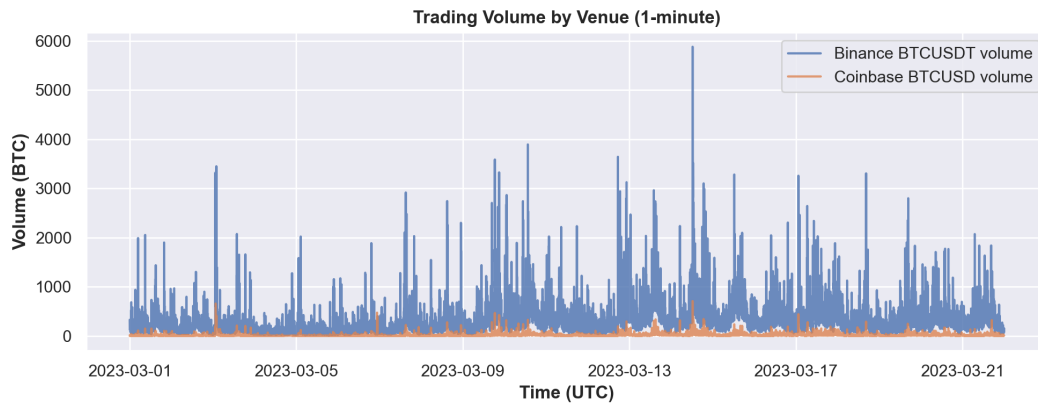


Figure 5: Trading volume on Binance (BTCUSDT) and Coinbase (BTCUSD). Volume surges on both venues during the March 10–13 stress window, but Binance dominates throughput by roughly an order of magnitude, underpinning its larger Hasbrouck information share.

3.4 Liquidity fragmentation across exchanges

[To be completed by team.]

4. Regulatory Overlay & Market Structure Implications

- **Stablecoin credibility drives the basis.** The USDC de-peg episode demonstrates that when a major stablecoin’s dollar peg is questioned, the cross-currency basis widens sharply. BTCUSDT traded at a persistent discount to BTCUSD because USDT inherited “contagion doubt” even though Tether itself maintained its peg—the market priced all crypto-dollars at a discount to bank USD.
- **Reserve backing and convertibility matter.** The one-sided nature of cost-adjusted deviations (exclusively USDT discount, zero positive breaches) suggests that the basis reflects a funding-risk premium tied to stablecoin reserve quality and redemption certainty. Regulation mandating transparent, segregated reserves would reduce this premium.
- **Institutional settlement implications.** The 7.5-hour half-life of basis dislocations implies that arbitrageurs face meaningful capital-at-risk windows. Faster, regulated settlement rails (e.g., stablecoin integration into payment systems under the GENIUS Act) would compress these windows and improve cross-venue price efficiency.
- **Efficiency vs. systemic risk.** The basis “bends but does not break”—the reversion rate ϕ is regime-invariant even as variance explodes $5.6\times$. This suggests the arbitrage mechanism is structurally sound but

capital-constrained. Regulatory frameworks that lower barriers to cross-venue capital flow would reduce dislocation magnitudes without disrupting the underlying price-discovery process.

- **Regulated stablecoins and trading patterns.** If the GENIUS Act establishes a credible regulatory floor for stablecoin solvency, the distinction between “crypto dollars” and bank USD diminishes. This would narrow the cross-currency basis in normal times and reduce the amplitude of stress-episode blowouts, converging toward FX-market efficiency benchmarks.

5. Implications for Market Participants

- **Arbitrage considerations.** Cost-adjusted basis exceedances of 13.3% (at $\tau = 0.50\%$) are economically meaningful, but the 7.5-hour half-life means arbitrageurs require significant capital duration. Execution must account for cross-venue transfer latency, fee asymmetry (Binance 10 bps vs. Coinbase 40 bps), and the risk that dislocations widen before reverting.
- **Venue selection.** Binance contributes 60% of efficient-price variance (Hasbrouck IS), making it the dominant price-discovery venue. However, Coinbase adjusts less rapidly to equilibrium, serving as the fiat-anchored reference. Informed traders may prefer Binance for speed; hedgers and compliance-sensitive participants may prefer Coinbase for USD settlement finality.
- **Treasury and risk management.** Stablecoin-quoted positions carry implicit funding risk that materializes during stress. The $5.6\times$ variance explosion and one-sided basis direction during March 10–13 imply that treasuries holding USDT-denominated BTC positions face asymmetric drawdown risk relative to USD-denominated equivalents. Hedging via basis monitoring or maintaining dual-venue presence reduces this exposure.